

- 11 The table shows assembly language instructions for a processor that has one register, the Accumulator (ACC).

| Label                                     | Instruction |           | Explanation                                                                                          |
|-------------------------------------------|-------------|-----------|------------------------------------------------------------------------------------------------------|
|                                           | Opcode      | Operand   |                                                                                                      |
|                                           | LDM         | #n        | Load the number n to the ACC                                                                         |
|                                           | LDD         | <address> | Load the contents of the location at the given address to ACC                                        |
|                                           | LDI         | <address> | The address to be used is at the given address. Load the contents of this second address to the ACC. |
|                                           | ADD         | <address> | Add the contents of the given address to the ACC                                                     |
|                                           | SUB         | <address> | Subtract the contents of the given address from the ACC                                              |
|                                           | STO         | <address> | Store the contents of the ACC at the given address                                                   |
| <label>:                                  |             | <data>    | Gives a symbolic address <label> to the memory location with the contents <data>                     |
| # denotes a denary number, e.g. #123      |             |           |                                                                                                      |
| <label> can be used in place of <address> |             |           |                                                                                                      |

The current contents of memory are:

| Address | Contents |
|---------|----------|
| 150     | 26       |
| 300     | 86       |
| 420     | 150      |

